

PalmSource®

Conference and Expo



Designing Compatible Applications

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As Palm OS® enters its fifth generation, compatibility is now more important than ever.



Agenda

- Benefits of Being Compatible
- Building In Compatibility
- Verifying Compatibility
- Common Problems + Solutions
- Common Misconceptions



What I Won't Cover Here

- Compatibility among various licensee devices
  - Come to session 201, Thursday 1:30-2:30!
- Anything that runs on the desktop
- Airspeed velocity of an unladen swallow
  - African
  - European



Benefits of Being Compatible

- Less work tomorrow
- Run on more devices
  - More potential customers = more sales
- Run on future devices
  - Happier customers (when they upgrade devices)
  - Happy customers = loyal customers



Palm OS® Versions

- With each new Palm OS release:
  - New features
  - Unsupported things often stop working

- Why support multiple Palm OS versions?
  - Probably the same reason you chose Palm OS in the first place: A broader market for your software



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Building In Compatibility

Build it in, don't "test it in"

- Compile against the latest SDK
- Disable structure access
- Check for OS support before using features
- Check return values!
  - Code defensively with the **`ErrFatalDisplay()`** macros



Forward Compatibility

- Do what's supported
  - If it's not supported, it might not work forever
- Don't do anything else
- If you do break the rules, expect to fail eventually and do so gracefully
  - Check those features
  - Check the processor type
  - Don't check device type
    - (it's probably not what you really want to know)



Recipe for Disaster

- Assume the presence of all the new features
- Break all the rules, assuming that things will never change.
  - Sure, the debug ROMs might complain, but it works on a real device, right?

This virtually guarantees that the software won't run on older OR newer devices.

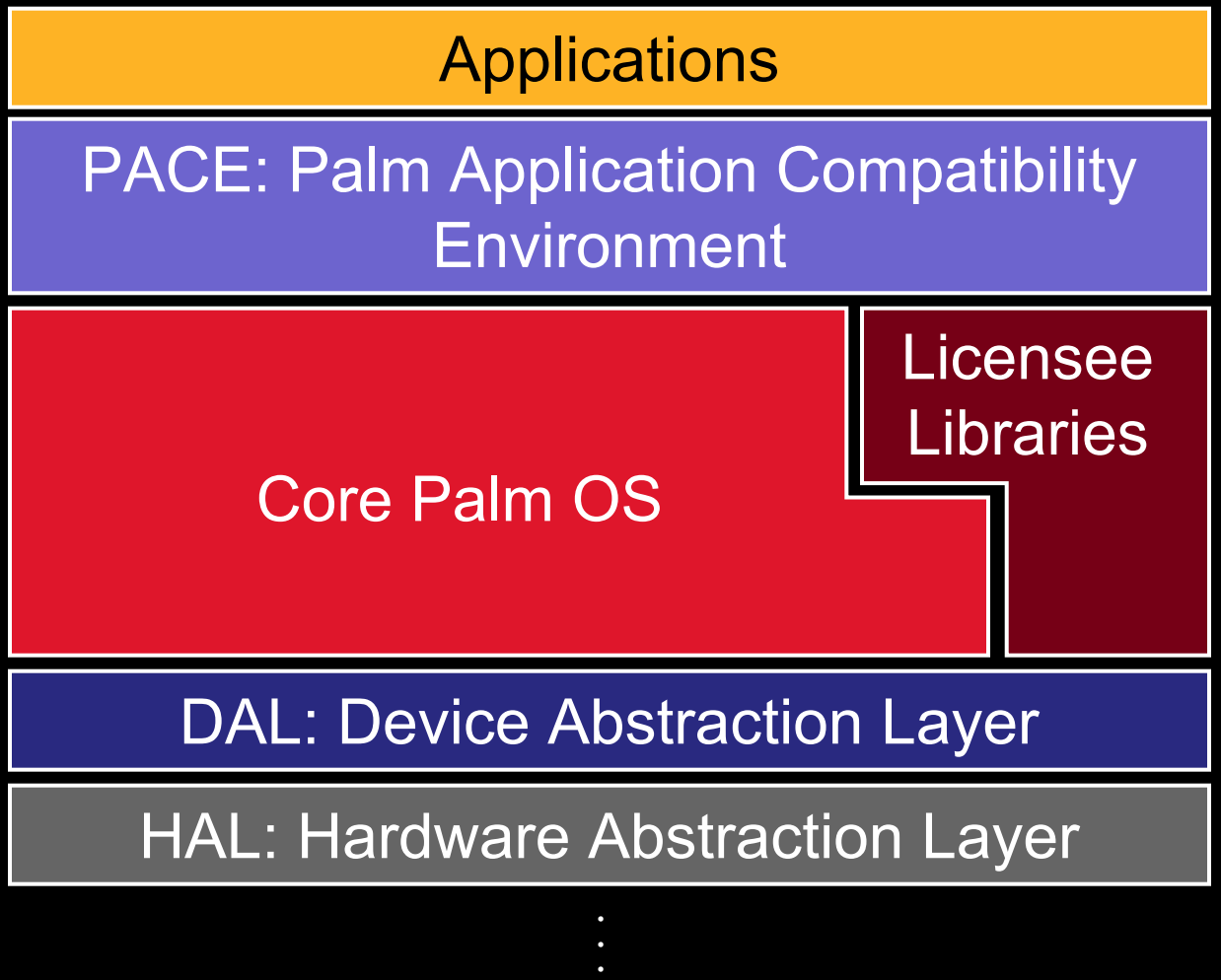


Forward Compatibility – PACE

Palm Application Compatibility Environment

- Performs 68k emulation
- Handles native OS/68k interaction
  - Endianness issues
  - Structure layouts
  - Shadow structures
  - Resource formats
  - Shadow databases
- Implements 68k debugger nub (you can debug just as though using Palm OS Emulator)

Palm OS 5 Block Diagram



Forward Compatibility—ARM

- Direct access to most UI structures is going away (we tried to warn you)
- Speed Compatibility
  - Time to start checking ticks
- Direct access to the display is more dangerous
  - New devices are using a greater variety of hardware
  - RGB ordering, endianness, bit-endianness
  - There's good news, too (more on this later)



Forward Compatibility Demo

We're Not Doing This Just to Be Mean

- Internals of structures change out of necessity
  - Endianness
  - UI is evolving to become more extensible and OO
  - This is why they're opaque
- There are lots of different ARM chips and display controllers
- PACE actually does a lot of work to maintain compatibility…
  - Shadow constructs
- …but it can't and shouldn't accommodate every bad habit

What Does *Opaque* Mean?

- For starters, don't do this:

```
controlP->font = boldFont;
```

- Or this:

```
ControlType tempControl;
```

- Or this:

```
... = sizeof(ControlType);
```



Why Is Backward Compatibility Important?

- How many ARM-based Palm OS devices are on the market today?
- How many 68k-based Palm OS devices are on the market today?



Backward Compatibility

- Check for features older system might not have
- If a feature is unavailable, try to run with that feature disabled
  - Much better user experience than not running at all
- Link in PalmOSGlue to bring new features to old devices



PalmOSGlue

- Included in the SDK
  - `#include "PalmOSGlue.h"`
  - Add the .lib to your project
- Two basic kinds of glue:
  - Functional
    - Enables new OS features on earlier OS versions
    - Example: `DateGlueToDOWDMFormat()`
  - Accessor
    - Provides a safe means to access UI structure elements
    - Example: `CtlGlueSetFont()`



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Verifying Compatibility

- Compile with the #define
  - This is the default in SDK 4.0 Update 1
- Palm OS Emulator and latest debug ROMs are a great start
  - Turn on all the warnings!
  - Emulator won't catch everything, yet
  - Everything it does warn about is almost certainly a bug (if not today, tomorrow)
- Palm OS Simulator



Introducing Palm OS Simulator

- Provides PACE environment on native *Windows*
- *Not* an ARM emulator
- *Not* related to Palm OS Emulator
- *Not* related to original Mac-based simulator
- Permits 68k debugging just like Emulator

What's So Great About Debug ROMs?

- Extra debugging assertions
- Lots of extra code to do pedantic validation of pointers, structures
- Debug color scheme by default
  - Reveals potential color scheme problems, overlapping UI, etc.



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Common Problems and Solutions

- Most common UI structure accesses:
  - Rectangle from field object
  - numItems from list object
  - Bounds from control object
  - attr from field object
- More reads than writes
- Solution: Use PalmOSGlue



Common Problems and Solutions, cont.

- Bad memory reads and writes
  - Low memory (for example, 0x00000000)
    - Classic return value problem
  - System globals
  - Unallocated chunks

- Solution: Use the APIs, check return values, validate pointers



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Common Misconceptions

- I need to write to the display directly
  - Try an offscreen window + WinCopyRectangle()
  - The blitter has been greatly optimized
  - The blitter is now running native ARM instructions!
- I need to patch traps
  - System notifications now make the vast majority of hacks unnecessary



Session Q&A



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